

JAEHYUN LEE

Rm 604, Bldg. K, 94, Wausan-ro
Mapo-gu, Seoul, South Korea 04066
+82 10-3741-5945
ljh00228@g.hongik.ac.kr
<https://ljh00228.github.io/jaehyunlee/>

EDUCATION

HONGIK UNIVERSITY

- M.S. in Materials Science and Engineering

Seoul, South Korea

Expected Mar 2027

HONGIK UNIVERSITY

- B.S. in Materials Science and Engineering
- Thesis title: Strain-Engineered and H₂O₂-Etched MoS₂ for Hydrogen Evolution Reaction

Seoul, South Korea

Mar 2025

RESEARCH

EXPERIENCE

HONGIK UNIVERSITY

1Q Lab

Advisor: Prof. Won-kyu Lee ([Link](#))

Graduate Research Assistant

Undergraduate research intern

Seoul, South Korea

Apr 2023 - Present

Mar 2025 – Present

Apr 2023 – Feb 2025

Materials Science and Engineering Researcher specializing in surface and interfacial science, nanoscale electrocatalyst design, and high-efficiency electrochemical energy-conversion devices.

- **Investigated** nanoscale structure-property relationships in water electrolysis systems via advanced spectroscopy and microscopy.
- **Currently developing** ionomer-free catalyst architectures for high-efficiency and durable electrochemical energy conversion for water electrolysis.
- **Interested** in nanomaterials, surface/interface engineering, wettability design, and atomic-scale simulation.

Cooperative Center for Research Facilities

Instrument Operator & Analyst, X-ray photoelectron spectroscopy

Supervisor: Prof. Chanyoung Yoo ([Link](#)) and Prof. Dongwook Lee ([Link](#))

- Coordinated internal and external analytical measurement requests for material characterization.

Seoul, South Korea

Feb 2025 – Present

PUBLICATION

(Published)

†: Equal contribution

*: Corresponding author

H. Jeon,[†] H. J. Kwon,[†] **J. Lee**,[†] S. K. Han, H. Kim, J. Heo, J. Han, S. Shin, J. Park, M. K. Cho, D. J. Preston, I. S. Kim,^{*} M. Kim, and W.-K. Lee,^{*} “Strain-Enabled Band Structure Engineering in Layered PtSe₂ for Water Electrolysis under Ultralow Overpotential,” *ACS Nano* **2025**, 19, 9107-9120 (Featured on Hongik News) [DOI](#)

J. Han,[†] M. K. Nam,[†] S. Shin,[†] I. Cho, H. Jeon, J. Heo, **J. Lee**, M. K. Cho, D. J. Preston, M. C. Hersam, I. S. Kim,^{*} B. Shong,^{*} and W.-K. Lee,^{*} “Efficient and Stable Electrocatalytic Oxygen Evolution from MoTe_x/Ni(OH)₂ Heterostructures,” *Journal of the American Chemical Society* **2025**, 147, 41096–41108 (Featured on National Research Foundation of Korea News) [DOI](#)

S. Shin,[†] I. Cho,[†] S. K. Han,[†] J. Heo, J. Han, H. Jeon, **J. Lee**, M. K. Cho, D. J. Preston, I. S. Kim,^{*} B. Shong,^{*} and W.-K. Lee,^{*} “Strain-Engineered Oxygen-Modified Nickel Telluride/Nickel Oxide Heterostructures for Bifunctional Alkaline Water Electrocatalysis,” *ACS Nano* **2025**, 19, 51, 42796–42815 (Back Cover article) [DOI](#)

S. Kang,[†] J. Pyo,[†] M. Jung, S. Park, T. Kim, H. Jeon, **J. Lee**, W.-K. Lee, Y.-W. Lee,^{*} S. Cha,^{*} and S. Pak,^{*} “Tunable Bandgap of Copper Sulfide via Cation Exchange as a p-Type Transparent Electrode,” *Nanoscale* **2026**, 18, 37, 20335–20346. [DOI](#)

S. K. Han,[†] Y. Kweon,[†] H. Jeon,[†] J. H. Cho, I. Kim, J.-H. Yim, **J. Lee**, J. Heo, S. Shin, S. J. Baek, M. K. Cho, Y. J. Jang, S. Y. Kim, I. S. Kim,^{*} M. Kim,^{*} and W.-K. Lee,^{*} “Amorphized and Oxidized Layered PtSe₂ as an All-in-One Electrocatalytic Platform for Hydrazine-Assisted Water Splitting,” *Advanced Materials* **2026**, e75207. (Featured on National Research Foundation of Korea News) [DOI](#)

(Submitted)

J. Lee,[†] S. Shin,[†] H. Jeon,[†] D. Lee, T. Zhou, I. S. Kim,^{*} and W.-K. Lee,^{*} “Beyond Monolayers: Activating Multilayer and Bulk Transition Metal Chalcogenides for Water Electrolysis,” *Materials Horizons* **2026**

(In preparation)

J. Lee, Y. Kang, and W.-K. Lee,^{*} “Networked Pt Clusterization on Mechanically Exfoliated PtSe₂ as an Ultra-Low-Pt-Loading Platform for Proton-Exchange Membrane Electrolyzer,” **2026**

CONFERENCES

J. Lee, H. Jeon, S. Han, J. Han, S. Shin, J. Park, and W.-K. Lee “Strain-Driven Electronic Modulation of Layered PtSe₂ for Advanced Hydrogen Evolution Catalysis,” 2025 *MRS Fall Meeting & Exhibit*, poster **2025, December**, Boston, USA. [DOI](#)

SKILLS

SURFACE CHARACTERIZATION

- X-ray and Ultraviolet Photoelectron Spectroscopy (XPS and UPS)
- X-ray Diffraction (XRD)
- Ultraviolet-Visible Spectroscopy
- Raman and FT-IR Spectroscopy
- In situ and operando characterization
- Scanning Electron Microscopy (SEM)
- Transmission Electron Microscopy (TEM)
- Atomic Force Microscopy (AFM)
- Contact Angle Measurements

MATERIALS TUNING

- Ar, N₂, O₂ Plasma-Assisted Etching
- Elastomer-Utilized Strain Engineering
- Electrochemical Dissolution and Oxidation
- Calcination, Sintering and Annealing
- Tip/Bath Sonication

DEVICE FABRICATION

- Membrane Electrode Assembly Fabrication and Characterization
- Dry/Wet Catalyst Loading on Polymer Membrane
- Nanoimprinting and Moulding

ELECTROCHEMICAL ANALYSES

- Potentio/Galvanostatic Operation
- Three/Two-Electrode Cell Testing
- Linear Sweep Voltammetry (LSV) and Tafel Analyses
- Electrochemical Impedance Spectroscopy (EIS)
- Cyclic Voltammetry (CV)
- Redox Characterization
- Liquid and Gas Chromatography
- In Situ Flow-Cell Characterization

MATERIALS SYNTHESIS

- Hydrothermal Synthesis of Reduced Graphene Oxide and Copper Sulfide
- Low-Pressure Chemical Vapor Deposition (LPCVD)
- Thermal Evaporation
- RF/DC Thin Film Sputtering
- Electrodeposition, Electroplating and Electrowinning
- Spin-Coating and Polishing
- Catalyst-Ink Fabrication

PROGRAMMING & LANGUAGE

- MATLAB, Python, Blender
- Korean (Native), English (Native Proficiency), Japanese (Intermediate), Chinese (Conversational)

AWARDS	Academic Excellence Scholarship	Spring 2024
	Undergraduate Research Assistantship Scholarship	Mar 2024 – Mar 2025
	National Science & Technology Scholarship	Mar 2025 – Mar 2027
ACADEMIC PROJECTS	Patent Landscape Analysis: Chiplet Packaging	Spring 2023
	Campus Patent Universiade Competition	
	Advisor: Prof. Kiyong Lee (Link)	
	Conducted a comparative analysis of over 3,600 Korean and U.S. patents related to 2.5D, 3D, and hybrid chiplet packaging technologies, focusing on major applicants including Intel and TSMC; identified core patents, technological gaps, and IP/R&D strategies for legal dispute prevention and new-market development.	
	Strain-engineered and H₂O₂-etched MoS₂ for Hydrogen Evolution Reaction	
	Undergraduate Capstone Project	Spring 2024
	Engineered H₂O₂-induced sulfur vacancies and strain-mediated phase transitions in MoS₂ to construct 2H/1T heterointerfaces with interfacial charge redistribution and enriched catalytic active sites; validated the material through XPS and electrochemical characterization, achieving HER activity approaching that of benchmark Pt.	
EXPERIENCES	REPUBLIC OF KOREA/UNITED STATES ARMY	Pyeongtaek, South Korea
	94th Military Police Battalion, 8th Army, United States Forces Korea	
	Military Police Officer and Army Interpreter	Apr 2019 – Sep 2021
	Conducted joint security patrols and law-enforcement operations with U.S. Army Military Police and the Republic of Korea National Police.	
OUTREACH	South Africa Vastfontein Program	Pretoria, South Africa
	Student Mentor	Nov 2017
	Supported fundraising initiatives for children's daily needs and education.	
	Community Meal Program	Suwon, South Korea
	Volunteer	Oct 2020
	Assisted with meal preparation and distribution for underserved communities.	
REFERENCES	Prof. Won-Kyu Lee Link	Dr. In Soo Kim Link
	Department of Materials Science and Engineering, Hongik University	Center for Nanoscale Materials, Argonne National Laboratory
	Email: wklee@hongik.ac.kr	Email: isk@anl.gov
	Phone: +82 10-7319-6870	Phone: +82 10-9391-9974
	Prof. Minho Kim Link	Prof. Dongwook Lee Link
	Department of Applied Chemistry, Kyung Hee University	Department of Materials Science and Engineering, Hongik University
	Email: minho.kim@khu.ac.kr	Email: dongwook@hongik.ac.kr
	Phone: +82 10-4631-8346	Phone: +82 2-320-1623